THE INVERSE SYSTEM OF POWERS OF AN IDEAL IS PRO-ISOMORPHIC TO THE SYSTEM OF KOSZUL COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 29647-29652 (canonical tag 0G3M) and the counted proof is the proof environment at source lines 29654-29674, which immediately follows it and is its own proof.

Editorial note. The counted statement is formulated inside the situation of the section; that situation environment is delivered as the context span of this card, so the objects named in the statement can be read and the two labels occurring inside it resolve in the compiled document.

Editorial note. Because the situation environment is delivered as the context span, no navigation stub is needed for the situation label; the stubs of this card exist for the three lemmas cited, and their kinds are the type words appearing in the labels themselves, not the names of the environments in which the labels live.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

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Editorial note. The revision r002 corrects the boundaries of the context span: the situation environment opens at source line 26304 rather than 26305 and closes at line 26318, so the first assembly of revision r001 ended with a stray closing of that environment and was rejected; no mathematical text was changed.

Allowed standard prerequisites (editorial).

- The inverse system of powers of an ideal is pro-isomorphic to the system generated by the n-th powers of any finite generating set. used at proof lines 29654-29656. Replaces the system of powers by the system generated by the powers of the chosen generators, matching the situation.
- If an inverse system of distinguished triangles has pro-zero cones then the two outer systems are pro-isomorphic. invoked at proof lines 29662-29664. Reduces the statement to showing that the system of cones is pro-zero.
- For a bounded inverse system of complexes it suffices to check that the cohomology systems are pro-zero, and the vanishing of those systems was established in the earlier Koszul discussion using Noetherianity. used at

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proof lines 29664-29673. Supplies the pro-zero property of the cones and hence finishes the proof.

Source reference. Original lemma 0624: sources/stacks-more-algebra/pinned-original-source.tex, source lines 6612–6612. The reference occurs inside the situation environment that is delivered as the context span, where it is used to produce the inverse system of Koszul complexes; it is a pointer needed only to define the objects of the statement.

Source reference. Original lemma 0921: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27172–27172. The reference is invoked at proof lines 29670-29672 for the pro-zero property of the Koszul cohomology systems, which is where the Noetherian hypothesis enters; it is the last input and everything leading to it is inside the counted span.

Source reference. Original lemma 0G3C: sources/stacks-chow/tags.txt, source lines 20087–20087. The reference reduces the pro-isomorphism of the outer systems to pro-zero-ness of the cones, invoked at proof lines 29662-29664; the computations establishing that property are inside the counted span.

Source reference. Original lemma 0G3B: sources/stacks-chow/tags.txt, source lines 20086–20086. The reference lets pro-zero-ness of a bounded system of complexes be checked on cohomology, invoked at proof lines 29664-29667; the degree computations that follow are inside the counted span.

Situation 0.2. Let A be a ring. Let $I = (f_1, \ldots, f_r) \subset A$. Let $K_n^\bullet = K_\bullet(A, f_1^n, \ldots, f_r^n)$ be the Koszul complex on $f_1^n, \ldots, f_r^n$ viewed as a cochain complex in degrees $-r, -r+1, \ldots, 0$. Using the functoriality of Lemma 0624 we obtain an inverse system

$$\ldots \to K_3^\bullet \to K_2^\bullet \to K_1^\bullet$$

compatible with the inverse system $H^0(K_n^\bullet) = A/(f_1^n, \ldots, f_r^n)$ and compatible with the maps $A \to K_n^\bullet$.

Lemma 0.3. *In Situation 0.2 assume A is Noetherian. With notation as above, the inverse system (I^n) is pro-isomorphic in $D(A)$ to the inverse system $(I_n^\bullet)$.*

Proof. It is elementary to show that the inverse system I^n is pro-isomorphic to the inverse system $(f_1^n, \ldots, f_r^n)$ in the category of A-modules. Consider the inverse system of distinguished triangles

$$I_n^\bullet \to (f_1^n, \ldots, f_r^n) \to C_n^\bullet \to I_n^\bullet[1]$$

where $C_n^\bullet$ is the cone of the first arrow. By Derived Categories, Lemma 0G3C it suffices to show that the inverse system $C_n^\bullet$ is pro-zero. The complex $I_n^\bullet$ has nonzero terms only in degrees i with $-r+1 \leq i \leq 0$ hence $C_n^\bullet$ is bounded similarly. Thus by Derived Categories, Lemma 0G3B it suffices to show that $H^p(C_n^\bullet)$ is pro-zero. By the discussion above we have $H^p(C_n^\bullet) = H^p(K_n^\bullet)$ for $p \leq -1$ and $H^p(C_n^\bullet) = 0$ for $p \geq 0$. The fact that the inverse systems $H^p(K_n^\bullet)$ are pro-zero was shown in the proof of Lemma 0921 (and this is where the assumption that A is Noetherian is used). $\square$